COSMOS-Rover: An Interplanetary Exploration Simulation

Computer Graphics Project - Department of CSE

Kazi Tasrif
Roll: 2007053

Naveed Lihazi
Roll: 2007038

January 27, 2026

1 Project Introduction

1.1 Overview

COSMOS-Rover is an interactive 3D graphics application developed using the **OpenGL 3.3 Core Profile**. The project simulates an interplanetary exploration rover navigating diverse celestial bodies, including Earth, Mars, and the Moon. Unlike static visualizations, this project emphasizes dynamic physics and atmospheric rendering to provide an immersive student perspective of space exploration.

1.2 Key Features

- **Multi-Planet Environments:** Distinct terrain rendering and skybox configurations for Earth, Mars, and the Moon.
- **Hierarchical Vehicle Simulation:** A detailed rover model with articulated parts, where wheel rotation scales with movement speed and the sensor mast responds to user input.
- **Variable Physics:** Real-time adjustment of gravity constants and movement mechanics adapted to planetary environments.
- **Advanced Shading:** Implementation of custom fragment shaders to calculate Ambient, Diffuse, and Specular lighting using the Phong Lighting Model.

2 Tools and Libraries

The application is built using a modern C++ pipeline in VS Code:

- **GLFW:** Utilized for window management and handling responsive keyboard inputs (W, A, S, D) for rover control.
- **GLAD:** Manages modern OpenGL function pointers for runtime hardware compatibility.
- **GLM:** Handles linear algebra for constructing Model, View, and Projection matrices.

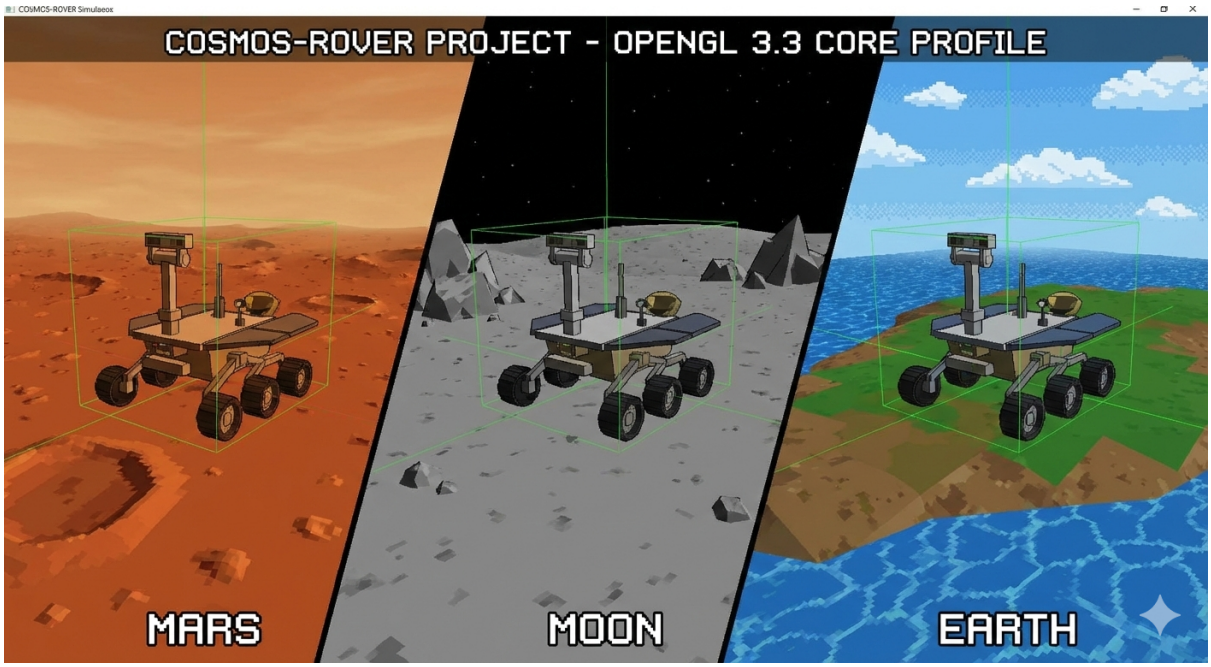

Figure 1: Simulated environments for Earth, Mars, and the Moon. (Image generated from Gemini)

3 Technical Implementation

3.1 Hierarchical Modeling Strategy

The COSMOS-Rover is constructed using a Scene Graph architecture to manage complex spatial relationships between its components. This hierarchical approach ensures that transformations propagate logically through the vehicle's structure:

- **Root Node (Chassis):** Acts as the primary coordinate frame. Any translation or rotation applied here moves the entire rover across the planetary terrain.
- **Wheel Child Nodes:** These nodes inherit the chassis's position but maintain independent local rotation around their X-axis. The rotation rate is procedurally animated to match the rover's linear velocity ($\theta = distance/radius$).
- **Sensor Mast Child Node:** Positioned atop the chassis, this node allows for independent Y-axis rotation. This enables the "scanning" animation and supports the First-Person camera orientation.

3.2 Multi-Perspective Camera System

To provide both immersion and navigational utility, the application utilizes a dynamic transformation pipeline that allows users to switch between three distinct viewing modes at runtime:

3.2.1 First-Person Perspective (POV)

In this mode, the camera is parented directly to the rover's Sensor Mast. By retrieving the mast's forward vector and applying a slight vertical offset, the simulation provides a "Driver's

Figure 2: Exploration Rover (Wall-E)

Perspective". This is ideal for detailed exploration of planetary surfaces and inspecting environmental textures.

3.2.2 Third-Person Follow View

This mode implements a "Chase Camera" that trails behind the rover. The camera position is calculated by subtracting the rover's forward vector from its current position and adding a height offset. This ensures a stable view of the rover's chassis while it navigates obstacles.

3.2.3 Strategic Top-Down View (Chase View)

For navigating complex craters or rocky Martian terrain, an elevated orthogonal or isometric perspective is provided.

- **Technical Goal:** The camera is positioned at a high Y-offset directly above the rover's root node.
- **Utility:** This "Bird's-Eye" view allows for strategic movement and clear visualization of the rover's Bounding Box relative to environment hazards.

3.3 Planetary Physics and Collision

The simulation adapts its physics engine based on the environment. We implement a gravity-based arc trajectory for the rover's suspension and jumping mechanics, using projectile motion

equations ($y = v_0 t - 0.5gt^2$) where g varies between Earth, Mars, and the Moon. Collision is managed via AABB (Axis-Aligned Bounding Box) logic, comparing the rover's extents against terrain obstacles to prevent mesh clipping.

4 Future Scope

Future enhancements include implementing **Shadow Mapping** for dynamic planetary shadows and a **Day-Night Cycle** to simulate the changing solar conditions on Mars and the Moon.